curve, the integral from $-\infty$ to $+\infty$, will always be equal to one. This is analogous to the sum of all of the pmf values being equal to one, and the sum of all of the histogram values being equal to *N*.

The histogram, pmf, and pdf are very similar concepts. Mathematicians always keep them straight, but you will frequently find them used interchangeably (and therefore, incorrectly) by many scientists and

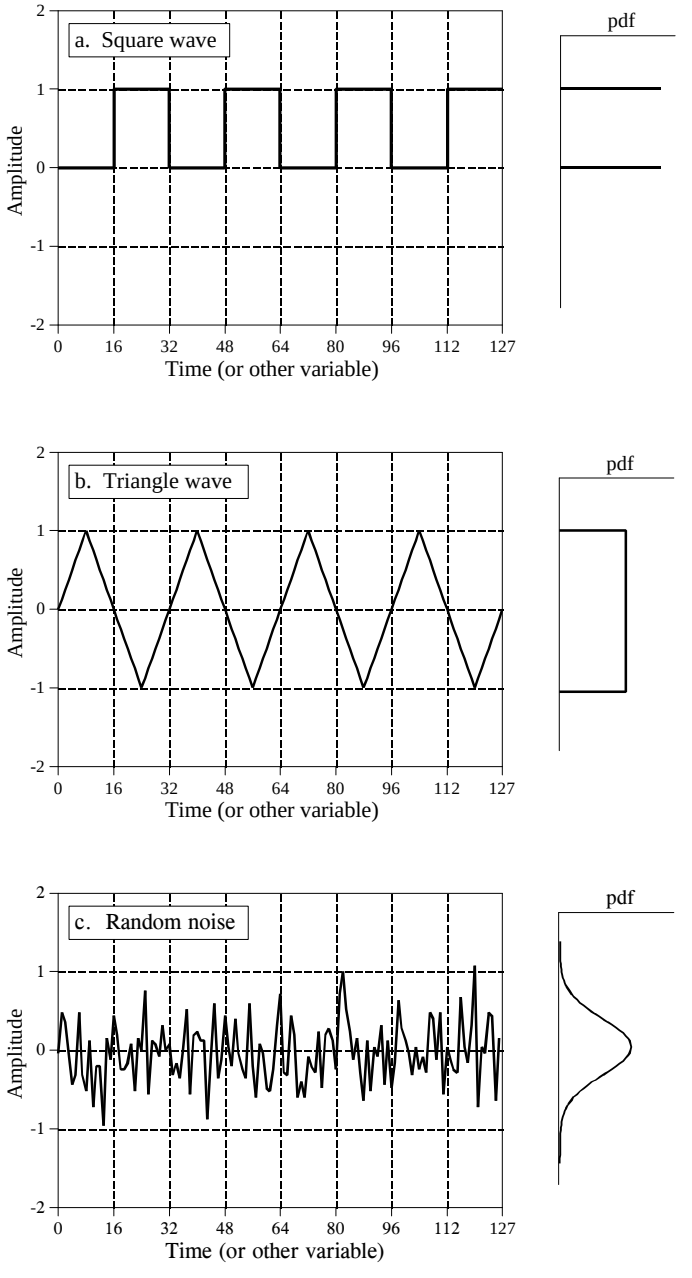

FIGURE 2-6
Three common waveforms and their probability density functions. As in these examples, the pdf graph is often rotated one-quarter turn and placed at the side of the signal it describes. The pdf of a square wave, shown in (a), consists of two infinitesimally narrow spikes, corresponding to the signal only having two possible values. The pdf of the triangle wave, (b), has a constant value over a range, and is often called a *uniform* distribution. The pdf of random noise, as in (c), is the most interesting of all, a bell shaped curve known as a *Gaussian*.